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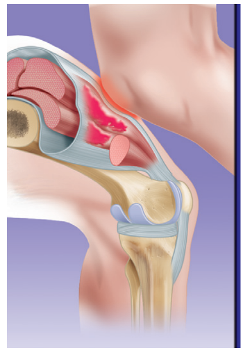


Injury Prevention

Jay Hertel, James Onate, Thomas Kaminski

Prevention of injuries is a critical, but often overlooked, component of the clinical practice of sports medicine. Incorporating injury prevention initiatives into an individual athlete's or team's training regimens requires forethought, planning, and persistence to maximize prophylactic benefits. To maximize these benefits, prevention initiatives must be targeted at known intrinsic or extrinsic injury risk factors, and targeted at athletes who are most susceptible to incurring the specific injuries being targeted for prevention. In this chapter, we highlight injury prevention initiatives aimed at three common lower extremity injuries: hamstring muscle strains, anterior cruciate ligament (ACL) ruptures, and ankle ligament sprains.

PREVENTION OF HAMSTRING MUSCLE STRAINS

Strains to the hamstring muscles are common occurrences in sports that involve maximal sprints and acceleration. Among sprinters, hamstring strains represent approximately one third of all acute injuries,[1] while they are also the first or second most common injury in soccer,[2–5] Australian rules football,[6,7] rugby,[8,9] and American football.[10,11] Hamstring strain injuries are also common in sports where the muscles may be stretched past the usual range of movement, such as in dancing and water-skiing.[13] There is evidence from American collegiate soccer that males are at a 64% increased risk of incurring hamstring strains than females.[14] Approximately 13% of all hamstring strains are recurrent injuries.[15]

CAUSES

Injury Mechanisms

The hamstring muscle group is composed of three muscles—semimembranosus, semitendinosus, and biceps femoris. All of them, except the short head of the biceps femoris, originate at the ischial tuberosity the pelvis. The two medial muscles, semimembranosus and semitendinosus, insert on the medial aspect of the proximal tibia, whereas the biceps femoris inserts onto the head of the fibula. The muscle group is biarticular, and its concentric actions are to extend the hip joint and flex the knee joint. However, it must be noted that the muscle group's eccentric actions in slowing hip flexion and knee extension are equally important.

Hamstring strains occur most often during maximal sprints. It is difficult to document exactly at what time injuries occur during the running cycle,[16] but the muscle group is thought to be vulnerable to strain injury in the late swing phase immediately before heel strike.[17,18] At this time point, the hamstring muscles work eccentrically.

RISK FACTORS

A number of risk factors have been proposed for hamstring strains, with the most prominent being the following four internal factors: reduced hip range of motion (ROM), poor hamstring strength, history of previous injury, and age.[19] In theory, limited ROM for hip flexion could mean that muscle tension is at its maximum when the muscle is vulnerable, at close to maximum length. However, this hypothesis has yet to be confirmed, because several studies involving soccer players suggest that hamstring flexibility is not a risk factor for strains.[20,21] However, other studies pertaining to soccer and Australian Rules football have shown that low quadriceps flexibility represents a risk factor not only for hamstrings[22] but also for quadriceps strains.[23]

Decreased hamstring strength would mean that the forces necessary to resist knee flexion and initiate hip extension during maximal sprints could surpass the tolerance of the muscle-tendon unit. Hamstring strength is often expressed relative to quadriceps strength as the hamstrings/quadriceps ratio, because it is the relationship between the ability of the quadriceps to generate speed and the capacity of the hamstrings to resist the resulting forces that is believed to be critical. Several studies show that players with decreased hamstrings/quadriceps ratios or side-to-side strength imbalances may be at increased risk of injury,[19] although the association has also been recently described as being of weak clinical importance.[24] Interestingly, reduced hamstring strength has been more strongly associated with increased risk of recurrent hamstring strains.[25]

A history of previous hamstring strains greatly increases injury risk.[19,26,27] Injury can cause scar tissue to form in the musculature, resulting in a less compliant area with increased risk of injury. A previous injury can also lead to reduced ROM or reduced strength, thereby indirectly affecting injury risk. Inadequate restoration of hamstring strength and endurance has been hypothesized as the primary reason for re-injury risk.[28]

Older players are at increased risk for hamstring strains, and although older players are more likely to have a previous injury, increased age is also an independent risk factor for injury.[21,27] Recent research has also provided preliminary evidence linking

Abstract

This chapter emphasizes four specific areas that are often the focus of injury prevention protocols: the hamstrings, spine, anterior cruciate ligament, and ankle.

Keywords

ankle
balance
taping
bracing
strength

Fig. 34.1 The Nordic hamstring exercise. Subjects are instructed to let themselves fall forward, and then resist the fall against the ground as long as possible by using their hamstrings. (Copyright Oslo Sports Trauma Research Center.)

an increased volume of high velocity sprinting with increased hamstring injury risk[29] and the congestion of multiple competitions in a short time period with overall injury risk.[30]

METHODS FOR PREVENTING HAMSTRING STRAINS

Research on injury prevention methods for hamstring strains is limited, and the evidence available has mainly been collected from observational studies. None of the prevention methods described here have been tested in large-scale randomized clinical trials with hamstring strains as the main end point. Studies to date have examined intervention methods targeting the key risk factors for hamstring strains: hamstring strength, hamstring flexibility, and previous injury.

The use of the Nordic hamstring exercise, emphasizing eccentric strengthening, has been shown to reduce the risk of hamstring strains.[2,31] This simple exercise is performed with a partner (Fig. 34.1) and starts with the athlete upright with their knees on the ground. While their partner holds the athlete's lower leg and feet in place, the athlete very slowly moves into bilateral knee extension as they lower their head, arms, and trunk to the ground. Surprisingly few sets and repetitions are needed to stimulate both a strengthening response and an injury prevention response.[2,8,31–34]

Because the Nordic hamstring lowers are easily implemented in a team setting, this exercise is recommended as a specific tool to prevent hamstring injuries. However, to avoid delayed-onset muscle soreness, it is important to follow the recommended exercise prescription with a gradual increase in training load when introducing a program of Nordic hamstring lowers (see Table 34.1).

The consistent finding that a history of previous injury leads to a several fold increase in the risk for new strains has led to the suggestion that this finding is at least partly due to inadequate rehabilitation and early return to sport. A study pertaining to Swedish soccer[34] has documented that a coach-controlled rehabilitation program consisting of information about risk factors

TABLE 34.1 Training Protocol for Nordic Hamstring Group

Week	Sessions/Week	Sets and Repetitions
1	1	2 × 5
2	2	2 × 6
3	3	3 × 6–8
4	3	3 × 8–10
5–10	3	3 sets, 12–10–8 repetitions

for reinjury, implementation of rehabilitation principles, and use of a 10-step progressive rehabilitation program, including return-to-play (RTP) criteria, reduced the re-injury risk by 75% for lower limb injuries in general. Although the specific effect on hamstring strains could not be assessed in this study, it seems reasonable to recommend the use of functional and specific rehabilitation programs and careful screening of players before RTP.

No intervention studies on the preventive effect of flexibility training on hamstring strains in elite athletes have been performed. However, one study involving military basic trainees indicated a reduced number of lower limb overuse injuries after a period of hamstring stretching,[35] whereas another military-based study found no effect of stretching.[36] It should be noted that these studies were designed to examine the effect of general stretching on lower limb injuries in general, not the effect of a specific hamstring program on hamstring strain risk. Questionnaire-based data on flexibility training methods collected from 30 English professional football clubs, where the stretching practices of the teams were correlated to their hamstrings strain rates, indicate that using a standard stretching protocol reduces injury risk.[37] Also, in one study from Australian Rules football, a reduction in the incidence of hamstring strains was observed with a three-component prevention program, with stretching while fatigued being one of the components.[38] The other factors in the program were sport-specific training drills and high-intensity anaerobic interval training. Because the program included three components, it is not possible to determine which of these factors are responsible for the observed effect.

In conclusion, the best evidence for hamstring injury prevention is available for programs designed to increase hamstring strength, particularly eccentric hamstring strength.

PREVENTION OF ANTERIOR CRUCIATE LIGAMENT TEARS IN THE FEMALE ATHLETE

One of the most debilitating types of lower extremity injuries is a rupture of the ACL. Surgical reconstruction costs, length of rehabilitation time, and risk for future long-term disability have generated significant interest for prevention programs aimed at reducing ACL injury risk. The number of ACL injury prevention programs has proliferated during the past 2 decades, with several research-investigated and clinically initiated programs being instituted worldwide.[39–58] The essential component of most of these programs is a multivariate approach aimed at creating adaptations in movement technique, neuromuscular control, fatigue resistance, injury awareness, and mobility/stability concepts aimed at altering high-risk movement patterns.[59–65] Most prevention programs institute components of injury risk awareness, strengthening of the lower extremity and trunk, enhancement of whole-body proprioception, increase in neuromuscular coordination, improvement of balance, and correction of potentially at-risk movement patterns through technique instruction and motor learning feedback systems in isolation or combination. Two recent position statements by US- and German-based groups reiterated the main programmatic paradigms for ACL injury prevention programs.[64,65] A 2018 position statement by the National Athletic Trainers' Association on Prevention of Anterior Cruciate Ligament Injury stated that a multicomponent ACL prevention program should at the minimum include three of the following exercise types: strength, plyometrics, agility, balance, and flexibility.[64] A 2018 report from the German Knee Society stressed the importance of screening, identification, and correction of endangering movement patterns as the first crucial steps to preventing ACL injuries.[65]

Implementation of an ACL injury prevention program is ultimately the responsibility of the entire sports medicine team, including the player, coach, parent, athletic trainer, physician, strength and conditioning specialist, physical therapist, and other persons as necessary.[59] The success of the program depends on an integration of roles for all persons involved and requires careful planning, execution, and review to evaluate the potential success of the program. Ideally, ACL prevention programs should be considered as primary injury prevention plans that are modifiable as the situation demands because of the clinical considerations of individuals/teams, given that various components of each program need to be personalized (e.g., soccer-specific vs. basketball-specific). Another consideration is the age of the participants for inducing biomechanical changes. A recent study from a Stanford University research group demonstrated that prevention programs created more significant biomechanical changes in preadolescent participants (aged 10 to 12) as compared with adolescent-aged (14 to 18) participants.[66] Long-term retention of proper movement patterns is an important component of ACL prevention programs and needs to be a critical consideration of all preventative efforts.[67] Careful planning and use of the Finch model of translating research into injury prevention practice should also be considered a cornerstone in evaluating the effectiveness of any ACL injury prevention program.[68]

Effective ACL injury prevention programs implement off-season, preseason, and in-season phases[43,50,52,58,59] to enhance and maintain individual gains. Most of the promising programs with increased compliance have a similar schedule, including sport-specific warm-up two to four times per week during the off-season, training two or three times per week during the preseason, and training one time per week during in-season practice.[39–59] The fundamental components of the most successful programs are dynamic warm-up, strengthening exercises, plyometrics, balance, sport-specific agilities, and motor learning focused feedback concerning proper movement techniques. An example of one program is the Federation International de Football Association Medical Assessment and Research Center (F-MARC) 11+ program (Figs. 34.2 and 34.3).[41] The F-MARC program requires

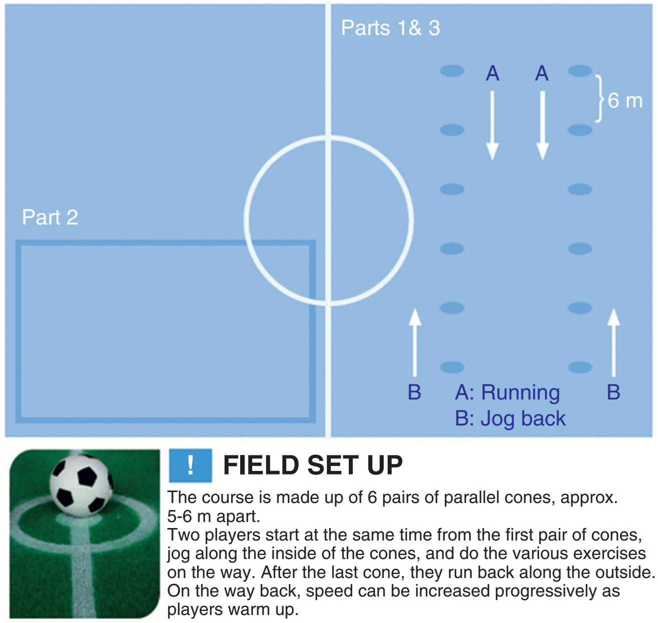


Fig. 34.2 Field setup. (From the Federation International de Football Association Medical Assessment and Research Center. More information is available at www.f-marc.com/11plus.)

minimal training and equipment (only a soccer ball required), making it an attractive alternative for persons with limited budgets and time. Three studies in which the F-MARC 11+ program was used with male and female adolescents demonstrated a reduction in injuries ranging between 21% and 71%.[46,56] The specificity of the 11+ soccer-focused training program can be performed in different sports by emphasizing the key components of the program and making the sport movement tasks match the desired activity. Similar training programs have been reported to produce a 49% reduction in lower extremity injury risk[46] and a 94% reduction in ACL injury risk[49] in Norwegian handball players.

Demonstration of program efficacy is essential before wide implementation. One of the main limitations to proving efficacy is that the "number needed to treat" to demonstrate a prophylactic effect as a result of training is relatively large. For pooled number needed to treat estimates, 89 persons (95% confidence intervals: 66 to 136) need to participate in the training program to prevent ACL injury during the course of a competitive athletic season.[62] This finding indicates that a typical youth recreational soccer program with a team of 20 participants would require approximately four teams to participate in the training program to prevent ACL injury. The cost/benefit of time and compliance aspects of programs must be considered when implementing a prevention training program. It is difficult to generalize findings from the various programs to different sports (e.g., soccer vs. team handball programs), different levels of play (e.g., youth vs. collegiate), varying program components (e.g., time of season, length of program, and amount of time per week), type of program (e.g., strength, balance, and feedback), and consistent integration across all levels of disciplines (e.g., strength and conditioning specialists, rehabilitation specialists, medical physicians, and sport-specific coaches).

Fig. 34.3 Federation International de Football Association (FIFA) Medical Assessment and Research Center 11 + program. (From the Federation International de Football Association Medical Assessment and Research Center. More information is available at www.f-marc.com/11plus.)

Most evidence demonstrates a moderate to strong level of evidence exists to support the implementation of ACL injury prevention programs to reduce the risk of ACL injuries.[60,61,63] A 2012 systematic review of the literature on the effectiveness of ACL injury prevention training programs[60] found that several studies[39,42,44,48,53–55] met their inclusion criteria for evaluating the effectiveness of ACL injury prevention programs. Most of these programs are directed at the young female athletic population, which is considered to be at the greatest risk for ACL injury. The investigators found that the pooled risk ratio was 0.38 (95% confidence interval [CI], 0.20 to 0.72) for the eight programs evaluated, resulting in a significant reduction in the risk of ACL rupture in the prevention group ($P = .003$), with a risk reduction of 52% in female athletes and 85% in male athletes.[60] A meta-analysis conducted by Donnell-Fink et al.[69] found 24 studies focused on knee and ACL injury prevention that consisted of 1093 participants across a time span consisting from 1996 to 2014. After the training interventions focused on neuromuscular and proprioceptive training exercises, the incidence ratio for knee injury was 0.731, and ACL injury was 0.493, thus indicating that a link existed between training programs and injury reduction. Even though therapeutic level II evidence moderately to strongly supports the effectiveness of ACL injury prevention training programs, caution is warranted because two sets of the studies (Mandelbaum et al.[48] and Gilchrist et al.[42]; Peterson et al.[53] and Peterson et al.[54]) report on essentially the same prevention programs at two different time frames. Nonetheless, the evidence in support of the effectiveness of ACL injury prevention programs is promising, with more research needed to confirm their effectiveness in various types of populations.

The possible use of a screening process to evaluate whether individual athletes truly need to be involved in an ACL injury prevention program should be considered, in addition to the evaluation of an individual athlete's post-ACL injury to ensure that he or she has undergone adequate progression to a safe stage of rehabilitation to tolerate such an intervention program. In addition, the retention aspects of movement pattern changes after intervention programs have yet to be conclusively validated; however, an initial report shows that a longer duration (9-month) program results in greater retention of movement pattern changes compared with a program of shorter duration (3 months).[52] Future research needs to focus on clinical screening tools to evaluate persons most in need of an intervention, program dosage, and retention effects of the intervention.

Attention to ACL injury prevention has grown steadily. The development and implementation of ACL injury prevention programs require a systematic methodologic approach. The bridging of the research-clinical gap to aid persons at risk for injury requires a multidisciplinary team using an evidence-based approach. Future randomized controlled trials and clinically based ACL injury prevention programs require a standardized database. Critical questions remain: specific biomechanical factors associated with increased ACL injury risk, modifiable factors related to increased ACL injury risk, vital components of ACL injury prevention programs, clinical approaches to accommodate for implementation of research-based evidence, and behavioral change models for organizational implementation need to be considered.

KEY POINTS TO CONSIDER FOR AN ANTERIOR CRUCIATE LIGAMENT INJURY PREVENTION PROGRAM

1. Because risk factors are multifactorial, the prevention strategies that need to be used are most likely multifactorial.
2. Prevention programs need to be tailored to meet the demands of the task and individual requirements.
3. More evidence is needed on prevention program implementation factors (e.g., supervision, time frame, and long-term motor learning effects).
4. No single program is best for all; a plan for prevention encompassing baseline screening and intervention must be developed by all members of the sports medicine team.
5. All solutions for ACL injury prevention have not been answered, but current evidence seems to show moderate effectiveness for several ACL injury prevention programs with minimal evidence showing detrimental effects of programs.

PREVENTION OF ANKLE SPRAINS

Ankle sprains are not only a public health burden; they also constitute the most common injury in sport. Fong et al. have suggested that the sports having the highest incidence of ankle sprains include field hockey, volleyball, football, cheerleading, ice hockey, lacrosse, soccer, gymnastics, softball, and track and field.[70] Prevention of ankle sprains both in those having sprain reoccurrences and in those who have never sprained is of importance to the clinician from both a time lost and cost standpoint. An evidence-based approach to injury prevention seems appropriate to guide the clinician in their efforts. In light of this, two recent peer-reviewed clinical guidelines[71,72] issued on the treatment of lateral ankle sprains in the athletic population will be used as a foundation for this section of the chapter on injury prevention.

There is no one single intervention that provides the best ankle sprain protection; instead it is wise for the clinician to adopt a multi-intervention strategy. The two facets of lateral ankle sprain prevention with the best evidence for use include the practice of taping and bracing, as well as a focused balance and neuromuscular control program.

Taping and Bracing

Both lace-up and semirigid ankle braces, along with traditional ankle taping, are effective in preventing ankle sprains and reducing the rate of reoccurrence in athletic populations. Although the mechanism of prevention is unclear, it is likely that taping and bracing provide mechanical stability and proprioceptive enhancement to the ankle joint. Interestingly, it does not appear that one is more effective than the other. Clinicians should therefore be guided by personal preference, practicality, and cost effectiveness. A recent report focusing on the benefits of both ankle taping and bracing suggests that bracing may be the best cost affordable alternative.[73] Despite being more than 40 years old, Garrick and Requa[74] present some of the best evidence available supporting the use of ankle taping to prevent ankle sprains; this was especially true in those with previous ankle injuries. There are numerous reports showing the effectiveness of ankle braces in reducing the incidence of ankle sprains in the athletic population.[75–78] Perhaps the most compelling case for ankle bracing comes from a recent tightly controlled randomized clinical trial by McGuine and colleagues,[78] who reported a significant reduction in ankle injuries among high school basketball players, regardless of previous injury. There is no evidence to support the notion that ankle taping and bracing increases the risk for injuries in other joints of the lower extremity kinetic chain (i.e., knee or hip). Furthermore, despite a recent recommendation in the clinical guideline released by the BJSM (2012) that recommends the use of taping and bracing be phased out over time, there is no evidence available to support this position.[71]

Balance and Coordination Training

Clinicians working with athletes should perform a multi-intervention prevention program, lasting at least 3 months, and one that is focused on balance and neuromuscular control to reduce the risk of ankle injury. This is especially true in those who have suffered previous ankle injuries. There is good support that enhancing and/or improving neuromuscular control translates into ankle joint injury reduction.[79] Of importance to the clinician is that there appears to be evidence that suggests both supervised and unsupervised balance coordination programs are successful in preventing ankle sprains.[80,81] This means that athletes do not necessarily have to receive supervised care in sports medicine facilities (clinics, athletic training rooms) to benefit from these types of intervention programs. Programs should include tasks involving single leg stance balance activities, balance training with perturbations (wobble board, balance mat), balance raining with simultaneous upper extremity movement, as well as dynamic jumping activities that incorporate balance and coordination. McKeon and colleagues[79] offer an excellent series of progressing balance and coordination activities that can be easily implemented in a clinic or at-home environment (Fig. 34.4).

Strength Training

Adequate muscle strength remains at the core of most successful athletic maneuvers and a cornerstone in ankle sprain rehabilitation. However, there is no direct evidence to suggest that strengthening in and of itself prevent ankle sprains. Intuitively, most clinicians would agree that strength in the musculotendinous structures in and around the ankle joint are necessary and play a role in providing both mechanical and dynamic stability to the ankle during athletic performance. Although the exact mechanism for this remains unclear, it is widely postulated that having an appropriate level of muscle strength helps correct imbalances that may put the foot/ankle into vulnerable positions relative to one's center of mass during athletic movements. Therefore interventions that serve to strengthen the lower leg (evertor, invertor, plantar flexors, and dorsiflexors) as well as the hip extensors and abductors should be an integral part of ankle sprain injury prevention protocols.

Dorsiflexion Range of Motion

Although there is no direct evidence supporting the use dorsiflexion ROM activities in ankle sprain prevention strategies, no one would argue against the need for appropriate ankle joint flexibility needed for athletes in all sport endeavors to function. Reductions in dorsiflexion ROM can lead to tightness of the gastroc/soleus complex and produce capsular adhesions that limit arthrokinematic motions in the ankle. There are reports of dorsiflexion deficits during dynamic jumping tasks[82] as well as during gait maneuvers[83] in populations with unstable ankles. The most appropriate theory is that the deficits prohibit the foot from attaining its full closed packed position (one of most support) during the stance phase of gait and in addition move the toes closer to the ground during the swing phase, placing the foot/ankle in a vulnerable position to roll over and sprain. With this in mind, it is prudent for the clinician to appropriately address dorsiflexion ROM deficits with interventions aimed at restoring the flexibility needed to carry out both activities of daily living, as well as sporting activities. These techniques should involve stretching exercises for the lower leg muscles and manual joint mobilization interventions that enhance ankle joint arthrokinematics.

Hypomobility Issues

Hypomobility at any joint in the lower extremity kinetic chain can challenge the motor-control mechanisms of the athlete and lead to joint instabilities.[84] It has been suggested that hypomobility occurs in the subtalar, talocrural, and the distal and proximal tibiofibular joints following ankle sprains. One might argue that considerations addressing the mobility of these joints prophylactically before a sprain event may be wise as well. Although there is no direct evidence supporting this type of clinical intervention strategy, clinicians should be aware of any signs of hypomobility and address them appropriately using joint mobilization techniques.[85]

Education of Risk

Educating athletes about the inherent risks of sport activities as well as vulnerable positions that may produce ankle sprains is an important role that all clinicians must take seriously. Athletes should be made aware of high-risk activities that may place the foot and ankle in a position of vulnerability and injury predisposition. Although there is no evidence supporting the practice of educating athletes concerning the risk for ankle sprain injury, it seems logical that this concept be used as a prevention strategy.

In summary, there is considerable support in the existing literature for the use of taping/bracing, as well as a multidimensional balance and coordination program to prevent lateral ankle sprains in the athletic population. Lesser support has been provided for interventions involving strength training, dorsiflexion ROM deficits, and risk education, although these should remain an integral part of an overall injury prevention program.

The Basics:

6 Categories of Exercises

1. Single-Limb Balance
2. Balance with Ball Toss
3. Balance and Reach
4. Fast Hopping
5. Hop to Stabilization
6. Hop to Stabilization and Reach

Progressions:
There are specific criteria for progression in each category
Progressions increase challenge of exercises in various ways, such as:
- Changing surface (firm to foam)
- Increasing duration of activity
- Increasing intensity of activity

Frequency of Training
3 times per week for 4 weeks
All sessions supervised by school ATC

Exercise 1: Single-Limb Balance
Objective: Stand as still as possible while standing on the involved limb.
Duration: 3 repetitions *Duration:* 3 repetitions
Progression:

Eyes Open	Eyes Closed
• Arms out for 15 seconds on firm surface	• Arms out on firm surface for 15 seconds
• Arms across chest for 15 seconds on firm surface	• Arms across chest on firm surface for 15 seconds
• Arms out for 30 seconds on firm surface	• Arms out on firm surface for 30 seconds
• Arms across chest for 30 seconds on firm surface	• Arms across chest on firm surface for 30 seconds
• Arms out for 15 seconds on foam surface	• Arms out on foam surface for 15 seconds
• Arms across chest for 15 seconds on foam surface	• Arms across chest for 15 seconds on foam surface
• Arms out for 30 seconds on foam surface	• Arms out for 30 seconds on foam surface
• Arms across chest for 30 seconds on foam surface	• Arms across chest for 30 seconds on foam surface
• Arms out for 60 seconds on foam surface	• Arms out for 60 seconds on foam surface
• Arms across chest for 60 seconds on foam surface	• Arms across chest for 60 seconds on foam surface

*All foam surface tasks will be performed on Airex pads.

Subjects cannot advance to next level in each category until they demonstrate 3 repetitions error free. Errors include:
1. Touching down with opposite limb
2. Excessive trunk motion (>30° lateral flexion)
3. Bracing the non-stance limb against the stance limb
4. Moving the stance foot from its starting position on the ground
5. Removal of arms from across chest during specified activities

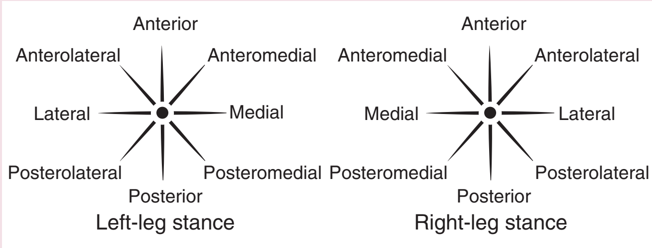
Exercise 2: Balance with Ball Toss
Object: Maintain prescribed stance while playing 2-handed catch with a 6-pound medicine ball.
Duration: 3 repetitions
Progression: Bipedal stance performed only until first stage of single-limb stance can be started.

Double-Limb Stance	Single-Limb Stance
5 throws in bipedal stance on firm surface (~20 seconds)	5 throws in unipedal stance on firm surface (~20 seconds)
10 throws in bipedal stance on firm surface (~40 seconds)	10 throws in unipedal stance on firm surface (~40 seconds)
15 throws in bipedal stance on firm surface (~60 seconds)	15 throws in unipedal stance on firm surface (~60 seconds)
5 throws in bipedal stance on foam surface (~20 seconds)	5 throws in unipedal stance on foam surface (~20 seconds)
10 throws in bipedal stance on foam surface (~40 seconds)	10 throws in unipedal stance on foam surface (~40 seconds)
15 throws in bipedal stance on foam surface (~60 seconds)	15 throws in unipedal stance on foam surface (~60 seconds)

Subjects cannot advance to next level in each category until they demonstrate 3 repetitions error free. Errors include:
1. Moving the stance foot (feet) from starting position on the ground
2. Excessive trunk motion (>30° lateral flexion)
3. Bracing the non-stance limb against the stance limb in single-limb stance
4. Failure to catch and throw the ball accurately back to clinician

Exercise 3: Balance and Reach
Object: Reach the specified distance in the prescribed direction with the uninvolved limb while maintaining balance on the involved limb. (Same principle as Star Excursion Balance Test – controlled movements).
Duration: 15 repetitions in each of 8 directions
Progression:

For each direction:
18″ allowed to use arms on firm surface
18″ hands on hips on firm surface
27″ allowed to use arms on firm surface
27″ hands on hips on firm surface
36″ allowed to use arms on firm surface
36″ hands on hips on firm surface
18″ allowed to use arms on foam surface
18″ hands on hips on foam surface
27″ allowed to use arms on foam surface
27″ hands on hips on foam surface
36″ allowed to use arms on foam surface
36″ hands on hips on foam surface



*Directions are named based on direction of reach in relation to the stance limb.

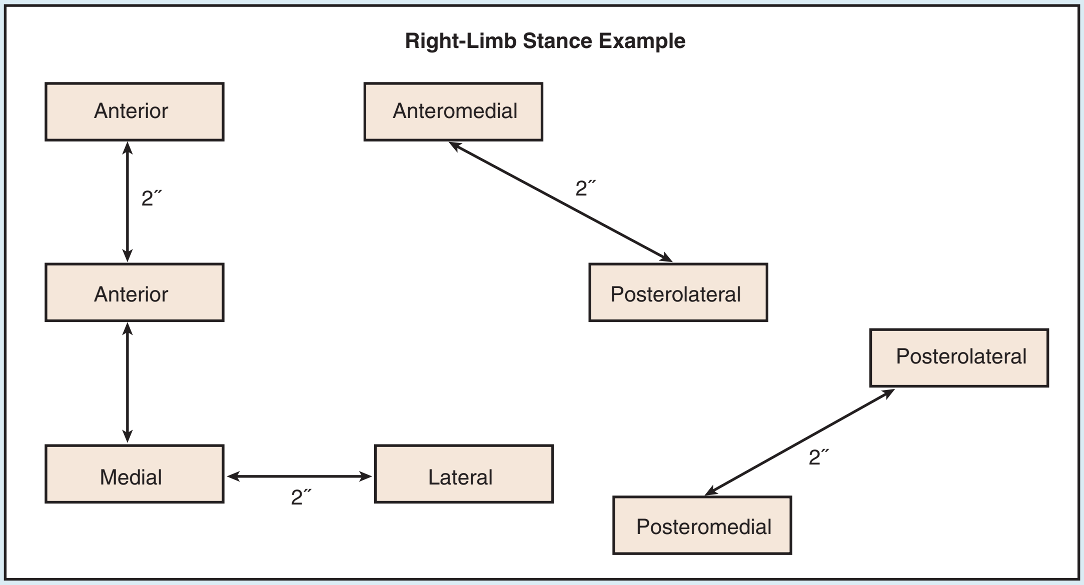
Subjects cannot advance to next level in each category until they demonstrate 20 repetitions error free. Errors include:
1. Touching down with reach limb other than at the intended target
2. Using the reach leg for a substantial amount of support during touchdown at end of reach
3. Moving the stance foot from its starting position on the ground
4. Bracing the reach limb against the stance limb
5. Removal of hands from hips during hands-on-hips activities

Exercise 4: Fast Hopping
Object: Hop in the prescribed direction, immediately hop back to the starting point, immediately repeat. Perform as quickly as possible. Involved limb only.
Duration: 3 sets of 5 repetitions in each of 4 directions (A/P, M/L, AM/PL, AL/PM)
Progression:

For each direction:
Hop across a 2″ line
Hop across a 4″ line
Hop across a 6″ line
Hop across a 9″ line
Hop across a 12″ line
Hop across a 6″ line and over 3″ high barrier
Hop across a 9″ line and over 3″ high barrier
Hop across a 9″ line and over 3″ high barrier

Subjects cannot advance to next level in each category until they demonstrate 3 sets of repetitions, all error free. Errors include:
1. Touching down with opposite limb
2. Not performing all hops continuously (i.e., prolonged stance)
3. Failing to jump completely across line or over barrier

Fig. 34.4 Balance intervention exercises. *ATC,* Certified athletic trainer. (Modified from McKeon PO, Ingersoll CD, Kerrigan DC, et al. Balance training improves function and postural control in those with chronic ankle instability. *Med Sci Sports Exerc*. 2008;40[10]:1810–1819.)



Exercise 5: Hop to Stabilization

Object: Hop to the target landing on the involved limb, stabilize and maintain balance in single-limb stance for 3 seconds, hop back to the starting point, again landing and stabilizing on involved limb for 3 seconds, repeat.

Duration: 3 sets of 5 repetitions in each of 4 directions (A/P, M/L, AM/PL, AL/PM)

Progression:

For each direction:
12″ hop allowed to use arms on firm surface
12″ hop hands on hips on firm surface
18″ hop allowed to use arms on firm surface
18″ hop hands on hips on firm surface
27″ hop allowed to use arms on firm surface
27″ hop hands on hips on firm surface
36″ hop allowed to use arms on firm surface
36″ hop hands on hips on firm surface
12″ hop allowed to use arms on foam surface
12″ hop hands on hips on foam surface
18″ hop allowed to use arms on foam surface
18″ hop hands on hips on foam surface
27″ hop allowed to use arms on foam surface
27″ hop hands on hips on foam surface
36″ hop allowed to use arms on foam surface
36″ hop hands on hips on foam surface

Subjects cannot advance to next level in each category until they demonstrate 3 sets of repetitions all error free. Errors include:

1. Touching down with non-stance limb
2. Bracing the non-stance limb against the stance limb
3. Excessive trunk motion (>30° lateral flexion)
4. Missing the target
5. Moving the stance foot from its landing position during stabilization
6. Removal of hands from hips during hands on hips activities

Exercise 6: Hop to Stabilization and Reach

Object: Hop to the target landing on the involved limb, stabilize yourself while maintaining balance in single-limb stance, reach with non-stance limb back to starting point (as in Exercise 3). Hop back to the starting point, again landing and stabilizing on involved limb, reach with non-stance limb back to previous target, repeat.

Duration: 3 sets of 5 repetitions in each of 4 directions (A/P, M/L, AM/PL, AL/PM)

Progression:

For each direction:
12″ allowed to use arms on firm surface
12″ hands on hips on firm surface
18″ allowed to use arms on firm surface
18″ hands on hips on firm surface
27″ allowed to use arms on firm surface
27″ hands on hips on firm surface
36″ allowed to use arms on firm surface
36″ hands on hips on firm surface
12″ allowed to use arms on foam surface
12″ hands on hips on foam surface
18″ allowed to use arms on foam surface
18″ hands on hips on foam surface
27″ allowed to use arms on foam surface
27″ hands on hips on foam surface
36″ allowed to use arms on foam surface
36″ hands on hips on foam surface

Subjects cannot advance to next level in each category until they demonstrate 3 sets of repetitions all error free. Errors include:

1. Touching down with reach limb other than for the prescribed reach
2. Bracing the non-stance limb against the stance limb
3. Excessive trunk motion (>30° lateral flexion)
4. Missing the target
5. Moving the stance foot from its landing position during stabilization
6. Removal of hands from hips during hands-on-hips activities

Fig. 34.4, cont'd

CONCLUSION

This chapter has highlighted the best practices for prevention of hamstring strains, ACL injuries in female athletes, and ankle sprains. Clinicians must use their best judgment when determining the need for implementing injury prevention initiatives with specific athletes and teams. It is critical that the sports medicine team consult with coaches and athletes about the best ways to incorporate injury prevention interventions into existing training programs to achieve maximum "buy-in" from all stakeholders.

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For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Goode AP, Reiman MP, Harris L, et al. Eccentric training for prevention of hamstring injuries may depend on intervention compliance: a systematic review and meta-analysis. *Br J Sports Med*. 2015;49(6):349–356.

Level of Evidence:
I

Summary:
This meta-analysis of clinical trials concluded that eccentric strengthening, with good compliance, was successful in prevention of hamstring injury.

Citation:
Kaminski TW, Hertel J, Amendola N, et al. National athletic trainers' association position statement: conservative management and prevention of ankle sprains in athletes. *J Athl Train*. 2013;48(4):528–545.

Level of Evidence:
I

Summary:
This expert consensus statement concluded that both lace-up and semirigid ankle braces and traditional ankle taping are effective in reducing the rate of recurrent ankle sprains in athletes, and that injury-prevention programs lasting at least 3 months focusing on balance and neuromuscular control reduce the risk of ankle injury in athletes.

Citation:
Padua DA, DiStefano LJ, Hewett TE, et al. National Athletic Trainers' Association position statement: prevention of anterior cruciate ligament injury. *J Athl Train*. 2018;53(1):5–19.

Level of Evidence:
I

Summary:
This expert consensus statement concluded that multicomponent injury-prevention training programs consisting of strength, plyometrics, agility, balance, and flexibility exercises are recommended for reducing noncontact and indirect-contact ACL injuries, as well as other knee injuries during physical activity.

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